**tsvc_c2/k53 · k53.c · full_nospeed_v4 · claude-haiku-4-5-20251001 · parallel-not-faster**

−2 / +13 lines of 13 · 2 parallel region(s) added · 6 of 15 candidate(s) kept · RESTRUCTURING (11 code line(s) added, 2 removed) · restructuring level 0

BEFORE

```
1  #include "data.h"
2
3  real_t kernel_k53(void)
4  {
5      for (int nl = 0; nl < iterations; nl++) {
6          for (long i = 1; i < LEN_1D; i++) {
7              u[ju[i]] += v[kv[i]] * c[i];
8              v[jv[i]] = u[ku[i]] * d[i] + c[i];
9          }
10         dummy(a, b, c, d, e);
11     }
12     return (real_t)0;
13 }
```

AFTER

```
1  #include "data.h"
2  #include <stdlib.h>
3
4  real_t kernel_k53(void)
5  {
6      for (int nl = 0; nl < iterations; nl++) {
7          real_t* v_local = (real_t*)malloc(sizeof(real_t) * LEN_1D);
8          #pragma omp parallel for shared(v_local)
9          for (long i = 0; i < LEN_1D; i++) {
10             v_local[i] = v[i];
11         }
12         for (long i = 1; i < LEN_1D; i++) {
13             u[ju[i]] += v_local[kv[i]] * c[i];
14             v_local[jv[i]] = u[ku[i]] * d[i] + c[i];
15         }
16         #pragma omp parallel for shared(v_local)
17         for (long i = 0; i < LEN_1D; i++) {
18             v[i] = v_local[i];
19         }
20         free(v_local);
21         dummy(a, b, c, d, e);
22     }
23     return (real_t)0;
24 }
```

file

red = removed  green = added  yellow = parallel region (#pragma omp)  right strip = the whole file: green where changed, yellow where a parallel region now is